| | |
|---|---|
| United States Patent Application Publication | 20250276369 |
| Kind Code | A1 |
| Publication Date | September 04, 2025 |
| Inventor(s) | Nelson; Scott et al. |


## RESIDUAL STRESS MANAGEMENT THROUGH CLOSED LOOP CONTROL


### Abstract

An additive manufacturing system includes an energy delivery device configured to deliver, a powder delivery device, and one or more thermal sensors configured to measure a temperature of a first portion of the additively-manufactured component and a second portion of the additively manufactured component. The additive manufacturing system includes a computing device configured to receive data indicative of the temperature of the first portion and of the second portion, determine a residual stress of the additively-manufactured component based at least partially on the received thermal sensor data from the first portion of the additively-manufactured component and the received data from the second portion of the additively-manufactured component; and predict final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.



**Inventors:** **Nelson; Scott (Indianapolis, IN), Puhl; David James (Indianapolis, IN), Grafton-Reed; Clive (London, GB), Daum; Peter E. (Indianapolis, IN), Proctor; Robert F. (Indianapolis, IN), Heason; Christopher Paul (London, GB)**

**Applicant:** **Rolls-Royce Corporation** (Indianapolis, IN); **Rolls-Royce plc** (London, GB)



**Family ID:** **96881625**

**Appl. No.:** **18/593344**

**Filed:** **March 01, 2024**


### Publication Classification

**Int. Cl.:** **B22F10/36** (20210101); **B22F10/28** (20210101); **B22F10/37** (20210101); **B22F12/90** (20210101); **B33Y30/00** (20150101); **B33Y50/02** (20150101)

**U.S. Cl.:**

CPC **B22F10/36** (20210101); **B22F10/37** (20210101); **B22F12/90** (20210101); **B33Y30/00** (20141201); **B33Y50/02** (20141201); B22F10/28 (20210101); B22F2203/11 (20130101)

## Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures.

SUMMARY

[0003] Additive manufacturing systems and techniques, such as directed energy deposition (DED) processes, operate according to a multifactor balance of mass flux and heat flux to deposit material layer-by-layer to form an additively-manufactured component (hereinafter, “component”). The material composition, geometry, and thermal history of the component may result in residual stress in the component. Residual stresses, such as those caused by different portions of the component thermally expanding and/or contracting at different rates, may cause deformation of the component such that one or more component dimensions change between an initial as-deposited state and a final state. Additive manufacturing systems and techniques according to the present disclosure may determine residual stress in the component in-situ (e.g., during the additive manufacturing process), and may use the determined residual stress to predict final component dimensions. In some examples, the predicted final component dimensions may be compared to planned final component dimensions (e.g. design dimensions), to validate that the component is being built correctly, or to stop the additive manufacturing process if the predicted final dimensions based on the residual stresses of the component do not meet a threshold for matching the planned final component dimensions. Further, systems and techniques according to this disclosure may modify a build strategy of the component during manufacturing to counteract or leverage the residual stresses in the component to result in a component where the actual final dimensions, after exertion of residual stress in the component, match the planned final dimensions.

[0004] An additive example manufacturing system includes an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, one or more thermal sensors configured to measure a temperature of a first portion of the additively-manufactured component and a second portion of the additively manufactured component, and a computing device. The computing device is configured to receive data indicative of the temperature of the first portion of the additively-manufactured component from the one or more thermal sensors captured at a first point in time and at a second point in time, receive data indicative of a temperature of the second portion of the plurality of portions of the additively-manufactured component from the one or more thermal sensors at a second point in time, determine a residual stress of the additively-manufactured component based at least partially on the received thermal sensor data from the first portion of the additively-manufactured component and the received data from the second portion of the additively-manufactured component, and predict final dimensions of the additively-manufactured component based at least

partially on the determined residual stress of the additively-manufactured component.

[0005] An example method includes receiving, by a computing device, data indicative of the temperature of a first portion of an additively-manufactured component from one or more thermal sensors of an additive manufacturing system captured at a first point in time and at a second point in time. The method further includes receiving, by the computing device, data indicative of a temperature of a second portion of the plurality of portions of the additively-manufactured component from the one or more thermal sensors at a second point in time. The method further includes determining, by the computing device, a residual stress of the additively-manufactured component based at least partially on the received thermal sensor data from the first portion of the additively-manufactured component and the received data from the second portion of the additively-manufactured component. The method also includes predicting final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.

[0006] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

---

## Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. **1** is a conceptual block diagram illustrating aspects of an example additive manufacturing system that includes a powder delivery device configured to direct a powder stream toward a melt pool a first energy delivery device, a second energy delivery device, and a computing device configured to control the additive manufacturing system during an additive manufacturing technique to determine, predict, and control residual stresses within the component.

[0008] FIG. **2** is a conceptual and schematic diagram illustrating further aspects of the additive manufacturing system of FIG. **1**.

[0009] FIG. **3** is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system.

[0010] FIG. **4** is an example calibration curve of particle detections versus mass flow.

[0011] FIG. **5** is a conceptual block diagram illustrating portions of the example additive manufacturing system of FIG. **1**, including an optical system for observing thermal emissions around a melt pool and a thermal camera for monitoring a size of the melt pool.

[0012] FIG. **6** is a conceptual block diagram illustrating an example optical system for observing thermal emissions around a melt pool formed during the additive manufacturing technique.

[0013] FIG. **7** is a process flow diagram illustrating a mass flux monitoring, heat flux monitoring, and residual stress prediction and control technique.

[0014] FIG. **8** is a conceptual and schematic diagram illustrating an additively-manufactured component in an initial as-deposited state.

[0015] FIG. **9** is a conceptual and schematic diagram illustrating an additively-manufactured component in a final state as distorted by residual stresses in the component.

[0016] FIG. **10** is a conceptual and schematic diagram illustrating an additively-manufactured component in an initial as-deposited state, accounting for predicted residual stresses in the component.

[0017] FIG. **11** is a flow diagram illustrating an example technique according to the present disclosure.

DETAILED DESCRIPTION

[0018] The disclosure generally describes techniques and systems for managing residual stresses during an additive manufacturing technique, such as a directed energy deposition (DED) technique.

During DED additive manufacturing, a component is built up by adding material to the component in sequential layers. The final component is composed of a plurality of layers of material. In some blown powder additive manufacturing techniques for forming components from metals, ceramics, and/or alloys, an energy source may direct energy at a substrate to form a melt pool. A powder delivery device may deliver a powder to the melt pool, where at least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate.

[0019] The properties of the final component, including the presence or absence of material defects, the resulting microstructure, and residual stresses within the component are a function of a number of variables related to mass flux and heat flux. As such, measurement, modeling, control, and validation of mass flux and heat flux within a blown powder additive manufacturing system (e.g., a DED system) may enable characterization or prediction of final component properties, control of the additive manufacturing technique during the process, quality assurance for the final component, development of new additive manufacturing techniques, and the like.

[0020] Challenges may arise while performing additive manufacturing techniques with additive manufacturing systems. For example, additively-manufactured components, especially those which include larger and/or thicker welds or a large number of deposited layers (e.g., hundreds of layers, or thousands of layers) may be subject to distortion/deformation caused by a residual stress or stresses within the component. The residual stress may strain the additively-manufactures component such that one or more dimensions of a layer or the component change between an initial as-deposited state to a final state where the component is cooled and ready for use. Residual stresses may be imparted to the component according to, among other factors, the thermal history of the component and the coefficient of thermal expansion of the materials.

[0021] For example, a cooling rate of the component may differ between a first portion of the component (e.g., located at an edge of the component) and a second portion of the component (e.g., located in the middle of the component) because thermal energy may diffuse more quickly from certain portions of the component to the environment. Thus, the first portion of the component and the second portion of the component may thermally expand and contract differently, leaving residual stresses in the component after manufacturing. Similarly, differences in material composition between portions of the additively manufactured component, or differences in topology between portions of the component, or the like may cause residual stresses to be present in the component. Residual stress, as used herein, may refer to any force acting on all or a portion of the component at any time immediately after formation of a portion of the component or the entire component by the additive manufacturing system.

[0022] Residual stresses may have deleterious impacts, including causing cracks or other weakness in the component or changing the desired geometry of the final component. For example, due to the residual stresses (e.g., forces of thermal contraction or thermal expansion) being exerted within the component, the component may deform by warping or bowing. Where the component geometry is critical (e.g., aerospace components such as gas turbine engine components including blades, tracks, shrouds, or the like), it may be important to understand and/or mitigate or eliminate residual stresses in the component to minimize distortion between the as-deposited dimensions of the component and the final dimensions after the component has cooled.

[0023] In accordance with one or more aspects of the current disclosure, additive manufacturing systems may address these and other problems. Additive manufacturing systems according to the present disclosure may include one or more thermal sensors configured to measure a temperature of a first portion of the component and a second portion of the additively manufactured component. The thermal sensor or sensors may include a thermal imaging camera or cameras and/or one or more temperature probes. The additive manufacturing system may include a computing device that receives data indicative of the temperature of a first portion of the component from the one or more thermal sensors captured at a first point in time and data indicative of the temperature of the first portion captured at a second point in time. The computing device may receive data indicative of a temperature of a second portion of the plurality of portions of the component from the one or more thermal sensors at a second point in time. The computing device may determine a residual stress of the component based at least partially on the received thermal sensor data from the first portion of the component and the received data from the second portion of the component.

[0024] In some examples, the computing device may predict the final dimensions of the additive manufacturing system based at least partially on residual stresses in the component in-situ, during manufacturing or after manufacturing without removing the component from a stage used to mechanically support the component during manufacturing. The determined residual stresses may be used to predict final component dimensions after the component has cooled and the distortion and deformation of the component has occurred according to the differing thermal contraction of different portions of the component. In some examples, if the predicted final dimensions of the component meet a threshold for matching the planned final component dimensions, the build may proceed according to plan. However, if the computing device determines that the residual stresses within the component will cause the component to deform such that the predicted final dimensions do not meet the threshold for matching the planned component dimensions, the computing device may output an alarm or may stop the build. As such, the disclosed systems may validate an additive manufacturing process while accounting for residual stresses in the component. Advantageously, the disclosed systems and techniques may result in reduced energy, material waste, and labor on components which will deform because of residual stresses and require rework or discarding.

[0025] In some examples, after determining that the predicted final dimensions of the component do not meet the threshold for matching the planned final dimensions, the computing device may control the additive manufacturing system based at least partially on the determined residual stresses to bring the component back into accordance with the specifications. For example, the additive manufacturing system may include a powder delivery device, a first energy delivery device, and a second energy delivery device. The computing device may control at least one of the powder delivery device, the first energy delivery device, or the second energy delivery device based at least partially on the predicted final dimensions of the component to adjust a build strategy of the component based on the determined residual stress. As discussed below, the build strategy may result in the final dimensions of the component. To control the first energy delivery device or the second energy delivery device, the computing device may modify at least one of a power, a travel speed, a spot size, or a power density of the first energy deliver device or the second energy delivery device. Additionally, or alternatively, the computing device may modify the first energy delivery device or the second energy delivery device by causing a pause time or dwell time, or may even cause a cooling device to remove thermal energy from one or more portions of the additively manufactured component. To control the powder delivery device, the computing device may modify the mass flow rate of the powder to adjust the as-deposited thickness of a layer being added to the component. In this way, the disclosed systems may prevent or manage residual stress in the component.

[0026] In some examples, residual stresses imparted in the component during manufacture may be desirable. For examples, a large ring weld, or a shrink-fit application, may benefit from deformation after manufacturing. In some examples, a mechanical bond between the component and other components may be improved by deformation after manufacturing due to residual stresses, similar to tightening opposite bolts in an assembly. The disclosed additive manufacturing systems, which may use deposit topology and the thermal history of the component collected by coaxial and off-axis sensors to build a high-fidelity predictive model of the development of the residual stresses of the component, may leverage the residual stresses in the component. For example, because understanding the residual stresses in the component may allow for understating the difference between the as-deposited dimensions of the component and the final dimensions of the component, the disclosed systems may allow for control of the system to deposit the as-deposited initial dimensions of the component differently than the planned dimensions of the component, such that the final dimensions of the component (e.g., after cooling and/or action of the residual stresses) are the planned dimensions of the component.

[0027] FIG. **1** is a conceptual block diagram illustrating aspects of an example additive manufacturing system **10**. Additive manufacturing system **10** includes several components configured to monitor mass flow, and several components configured to monitor energy (e.g., heat flux) within system **10**. System **10** includes a powder source mass sensor **44**, a powder flow monitoring system (PFMS) **18**, and a topology sensor **48**. These components are configured to monitor mass flow of powder within additive manufacturing system **10** during an additive manufacturing technique. In the example illustrated in FIG. **1**, additive manufacturing system **10** further includes a computing device **12**, a powder delivery device **14**, a first or primary energy delivery device **16**, a second or secondary energy delivery device **17**, a stage **20**, a powder source **42**, powder source mass sensor **44**, topology sensor **48**, and microstructural monitoring device (MMD) **19**. Computing device **12** is operably connected to powder delivery device **14**, energy delivery devices **16** and **17**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, topology sensor **48**, and MMD **19**. FIG. **1** thus illustrates mass flow monitoring and other aspects of example additive manufacturing system **10**. To simplify illustration of FIG. **1** and improve clarity of the figure, further aspects of additive manufacturing system **10** are shown in FIG. **5** and described below with reference to FIG. **5**, which is more directed toward heat flow aspects of system **10**.

[0028] Stage **20** is configured to mechanically support a component **22** during an additive manufacturing technique. Component **22** may be considered in-situ when mechanically supported by stage **20**. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. For example, stage **20** may be translatable and/or rotatable along at least one axis to position component **22** relative to energy delivery device **16** and/or powder delivery device **14**. Similarly, energy delivery device **16** and/or powder delivery device **14** may be translatable and/or rotatable along at least one axis to position energy delivery device **16** and/or powder delivery device **14**, respectively, relative to component **22**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0029] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream.

[0030] System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0031] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0032] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder. The combined powder defines powder stream **30**. In some examples, powder delivery device **14** includes a single nozzle, which may be point nozzle, or a single nozzle that is an annular channel. In other examples, powder delivery device **14** includes a plurality of nozzles (e.g., three nozzles or four nozzles). Regardless of the number of nozzles, powder delivery device **14** may output a powder stream that is focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. **1** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.

[0033] At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**. At least some of the powder that impacts melt pool **32** may be joined to component **22**. In some examples, powder delivery device **14** may be mechanically coupled or attached to primary energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**.

[0034] Primary energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0035] In some examples, primary energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**. Primary energy delivery device may be configured to focus energy **34** from the energy source on a local spot on build surface **28** to generate melt pool **32**.

[0036] In some examples, at least a portion of primary energy delivery device **16** and powder delivery device **14** may be combined or attached to each other. For example, a deposition head (e.g., deposition head **54** of FIG. **2**) may include part of powder delivery device **14** (e.g., internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**) and part of primary energy delivery device **16** (e.g., the energy delivery head). As shown in FIG. **1**, in some examples, primary energy delivery device **16** may be arranged of configured such that energy **34** and powder stream **30** both exit from a common deposition head (**54**, FIG. **2**) and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel (e.g., formed along central longitudinal axis L, FIG. **2**) within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**. Such an arrangement between primary energy source **16** and powder delivery device **14** may be called an “on-axis” arrangement of primary energy source **16**, because both energy and powder may be delivered coaxially with a central longitudinal (Z-direction) axis of the deposition head.

[0037] System **10** also includes powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light source. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28** (e.g., central longitudinal axis L).

[0038] The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. Because of the velocity of the powder in powder stream **30**, even such a frame rate may image only a fraction of the powder flowing between powder delivery device **14** and build surface **28**.

[0039] In some examples, PFMS **18** also includes a housing configured to enclose the illumination device and the imaging device. The housing may be configured to protect the illumination device and the imaging device from damage due to the harsh conditions to which PFMS **18** may be exposed during use. For example, the housing may protect the illumination device and the imaging device from powder deflections from powder stream **30** off build surface **28**, may cool the illumination device and the imaging device to remove heat incident on PFMS **18** from melt pool **32** and energy delivery device **16**, or the like.

[0040] PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16**, e.g., in the x-y plane shown in FIG. **1**. This may help maintain a relative x-y position of PFMS **18** and the image plane of the imaging device relative to powder stream **30**. This may facilitate analysis of image data captured by the imaging device.

[0041] PFMS **18** may be movable in the z-axis direction of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may enable movement of image plane **38** along the z-axis of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may allow PFMS **18** to image powder stream **30** at different positions between powder delivery device **14** and build surface **28**. In this way, PFMS **18** may analyze powder stream **30** along powder stream **30** to help determine parameters of powder stream **30** along its length.

[0042] In some example, PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16** and movable parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28** by an adjustable z-stage **40**. Adjustable z-stage **40** may be attached to energy delivery device **16**, powder delivery device **14**, or a portion of system **10** that moves energy delivery device **16** and/or powder delivery device **14**, such that PFMS **18** moves in the x-y axis in registration with energy delivery device **16** and/or powder delivery device **14**.

[0043] Adjustable z-stage **40** may be controlled by computing device **12** to position PFMS **18** and image plane **38** relative to powder stream **30**. Further, computing device **12** may control adjustable z-stage **40** to move PFMS **18** vertically and out of the way to allow powder delivery device **16** and energy delivery device **16** access to physically constrained areas, e.g., between vanes of a doublet or triplet of a nozzle guide vane for a gas turbine engine.

[0044] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to monitor an amount of powder captured by melt pool **32** by imaging melt pool **32** and the added material, allowing the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is a great as about 10 microns (e.g., about 6 microns).

[0045] In some examples, topology sensor **48** may be positioned substantially directly above component **22** and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses

depth information without using an interferometer.
[0046] In some examples, topology sensor **48** may be integral with system **10**, e.g., disposed within the enclosure or working area of system **10**. In other examples, topology sensor **48** may be an add-on component to system **10**. For example, the enclosure in which the additive manufacturing technique is performed may include a transparent window, and topology sensor **48** may be positioned outside of the enclosure and may image component **22** through the transparent window.
[0047] Although a topology sensor **48** is described in the examples of this disclosure, in other examples, another metrology device may be utilized to determine the amount of powder captured by melt pool **32**. For example, another type of light source may be used. In some examples, if another type of light source is used, component **22** or stage **20** may include one or more features that serve as indicators of scale. Furthermore, although described as a single topology sensor **48**, more than one sensor may be used, and may employ more than one of the technologies described above.
[0048] System **10** further includes secondary energy delivery device **17**. In some examples, secondary energy device **17** may include an energy source and energy delivery head as described above with respect to primary energy delivery device **16**. As such, secondary energy delivery device **17** may be configured to deliver energy **34** to a second spot on build surface **28**, as will be further described below. In some examples, secondary energy delivery device **17** may be displaced from a central longitudinal axis of the deposition head, and thus may be called an “off-axis” energy source. Alternatively, secondary energy delivery device may be configured to deliver energy globally to component **22** (e.g., the entire component body **22**). Furthermore, although it is considered that the energy source for secondary energy delivery device **17** may be a laser, other sources of energy are considered. For example, the energy source for secondary energy delivery device **17** may include one or more of an induction heater, an infrared heater, a microwave heater, a fan or blower system configured to deliver hot gases to build surface **28**, or the like.
[0049] System **10** may optionally include microstructure monitoring device (MMD) **19**. MMD **19** may be configured to capture data representative of the microstructure of component **22** in-situ, and output the captured data to computing device **12**. MMD **19** may include an imaging sensor such as an X-Ray device, a computed tomography (CT) device, a magnetic resonance imaging (MRI) device, or the like. Additionally, or alternatively, MMD **19** may include an acoustic sensing system such as an ultrasound device. Although illustrated as an off-axis add-on to system **10**, in some examples MMD **19** may be part of a deposition head and be arranged on-axis.
[0050] Computing device **12** may control components of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may control operation of system **10**, including, for example, powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, topology sensor **48**, and/or MMD **19**. Computing device **12** may be communicatively coupled to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** using respective communication connections. In some examples, the communication connections may include network links, such as Ethernet, ATM, or other network connections. Such connections may be wireless and/or wired connections. In other examples, the communication connections may include other types of device connections, such as USB, IEEE 1394, or the like.
[0051] Computing device **12** may include one or more processors. Example of processors include, but are not limited to, one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components.
[0052] Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include

multiple computing devices **12**, e.g., a plurality of computing devices **12**. In general, control and processing functions described herein may be divided among one or more computing devices. For instance, system **10** may include a controller for energy delivery devices **16** and **17**, powder delivery device **14**, and stage **20**, a separate controller for PFMS **18**, and a separate computing device for analyzing data obtained by PFMS **18**, mass sensor **44**, topology sensor **48**, and MMD **19**. As another example, system may include a dedicated controller for each of primary energy delivery device **16**, secondary energy delivery device **17**, powder delivery device **14**, stage **20**, PFMS **18**, topology sensor **48**, and MMD **19**, and a separate computing device for coordinating control of powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** and analyzing data obtained by PFMS **18** powder source mass sensor **44**, and/or topology sensor **48**. Other examples of computing system architectures for controlling system **10** and analyzing data obtained from system **10** will be apparent and are within the scope of this disclosure.

[0053] Computing device **12** may control operation of powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, adjustable z-stage **40**, stage **20**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, topology sensor **48**, and MMD **19**. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, adjustable z-stage **40**, topology sensor **48**, and/or MMD **19** to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery devices **16** and **17**, PFMS **18**, topology sensor **48**, and MMD **19**. Positioning component **22** relative to powder delivery device **14**, energy delivery devices **16** and **17**, PFMS **18**, topology sensor **48**, and MMD **19** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, topology sensor **48**, and/or MMD **19**.

[0054] Computing device **12** may control system **10** to deposit layers **24** and **26** to form component body **25** and eventually finished component **22**. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers. Furthermore, although component **22** is illustrated as being uniform, in some examples component **22** may be functionally-graded and include at least two different portions having different selectively tailored properties, as will be further illustrated and described below.

[0055] To form component **22**, computing device **12** may control powder delivery device **14**, primary energy delivery device **16**, and secondary energy delivery device **17** to form, on a surface **28** of first layer **24** of material, a second layer **26** of material using an additive manufacturing technique. Computing device **12** may control primary energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0056] Computing device **12** then may control a z-axis position of stage **20** and/or powder delivery device **14** and primary energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may control powder delivery device **14** and energy delivery device **16** similarly until all layers are formed to define a completed component **22**.

[0057] Computing device **12** may control powder delivery device **14** and primary energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may control secondary energy delivery device **17** to deliver energy to a portion of build surface **28** to modify the cooling rate, solidification behavior, or other aspects of the thermal profile of component **22** determined based on data captured by one or more thermal sensors (not illustrated in FIG. **1** for clarity) associated with portions of component **22**. The one or more thermal sensors may be configured to capture and store data during the build (e.g., at a first point in time, a second point in time, a third point in time, etc.) taken from a first portion of component **22**, a second portion of component **22**, and optionally additional portions of component **22**.

[0058] Computing device **12** may determine, based on the captured data by the one or more thermal sensors, a residual stress of all or a portion the additively-manufactured of component **22**. In some examples, to determine the residual stress, computing device **12** may calculate thermal expansion or contraction of component **22** based at least partially on the data captured by the one or more thermal sensors. In some examples, computing device **12** may calculate a cooling rate of the first portion or the second portion of the component **22** by comparing the data captured at the first point in time to the data captured at the second point in time. In some examples, computing device **12** may compare the cooling rate of the first portion to the cooling rate of the second portion to determine the residual stress of the component. In some examples, to determine the residual stress, the computing device is configured receive data indicative of a temperature of a plurality of portions of the component in addition to the first portion and the second portion.

[0059] Based partially or completely on the determined residual stress in component **22**, computing device **12** may predict final dimensions or update predicted final dimension of component **22**. Computing device **12** may then analyze the determined residual stress data to determine whether the predicted final dimensions of component **22** meets a threshold for matching planned or originally modeled dimensions of component **22**. For example, the threshold for matching planned final dimensions may mean that build surface **28**, and/or one or more dimensions of component **22** remains within 1 millimeter (mm), or.5 mm, or 0.1 mm, or 0.01 mm, or the like, of a modeled or planned build surface **28** or other component **22** throughout the build, and component contains residual stresses that will not cause build surface **28** or other component dimensions to warp, distort, or deform such that the component **22** no longer matches the planned dimensions.

[0060] Computing device **12** may perform one or more operations based on whether the predicted final dimensions of component **22** meets a threshold for matching planned or originally modeled dimensions of component **22**. As one example, upon determining that the predicted dimensions based at least partially of the determined residual stress, computing device **12** may output a warning to an operator, or computing device **12** may stop an additive manufacturing technique being performed by system **10**. In this way, system **10** may perform a closed feedback loop to avoid building component **22** that will deform due to residual stresses in component **22** such that that actual dimensions of component **22** do not match the planned dimensions of component **22** that were desired and designed before beginning the additive manufacturing procedure (e.g., the planned dimensions may be pre-determined dimensions, e.g., from a CAD model or the like).

[0061] As another example, computing device **12** may modify a build strategy. For instance, computing device **12** may be configured to control at least one of powder delivery device **14**, first energy delivery device **16**, or second energy delivery device **17** based at least partially on the predicted final dimensions of the component **22** to adjust a build strategy of the component based on the determined residual stress. As used herein, the build strategy is the combination of settings of system **10** that result in the final dimensions of the component **22**. To control first energy delivery device **16** and/or second energy delivery device **17**, computing device **12** may then modify one or more of the magnitude, power density, travel speed, or spot size (i.e., focus area) to generate a portion of component **22** (e.g., all or a portion of layer **26**) to selectively tailor the characteristics of the portion based on the determined residual stress of the portion. For example, computing device **12** may cause second energy delivery device **17** to pause, dwell, or skip certain portions of build surface **28** to modify the determined residual stresses of component **22** or of layer **26**. As one example computing device **12** may cause secondary energy delivery device **17** to add energy **34** to portions of component **22** that would cool more quickly (e.g., edges of component **22**), and thus generate residual stresses due to thermal contraction, than other portions (e.g., central portions of component **22**) of component **22**. Managing the heat flux of component **22** prior to, during, and/or subsequent to deposition of layer **26** may stabilize component **22** such that thermal expansion and contraction of different portions of component **22** (e.g., a first portion, a second portion, a third portion, etc.) is stabilized (e.g., occurs at the same or similar rate) relative to other portions of component **22**. In this way, residual stresses in component **22** may be limited or eliminated, and system **10** may mitigate or prevent deformation of component due to residual stresses in component **22**.

[0062] In some examples, in addition to or alternative to modification of energy **34** delivered by first energy delivery device **16** and/or second energy delivery device **17**, computing device **12** may modify mass flow to component **22** to counteract or cancel out predicted deformation of layer **26** and/or component **22**. For example, computing device **12** may manipulate the mass flow rate of powder in powder stream **30** delivered by powder delivery device **14** to adjust the as-deposited thickness of layer **26**. For example, responsive to determining that underlying layer of layers **24** of component **22** contain residual stresses that will cause deformation of component **22** is a particular direction (e.g., bowing), computing device **12** may cause layer **26** and optionally subsequent layers to be deposited such that layer **26** contains residual stresses that will cause component **22** to deform in the opposite direction. In some examples, the opposing residual stresses may cancel out, resulting in final component **22** that has actual dimensions that meet the threshold for matching the planned dimensions of component **22**. Additionally, or alternatively, computing device **12** may cause powder delivery device **14** to modify the mass flow rate of powder in powder stream **30** to deposit more or less material on or more portions of component **22** as layer **26** to account for the predicted final dimensions of component **22** to bring the predicted final dimensions of component **22** in line with the planned dimensions.

[0063] In some examples, computing device **12** may store/execute one or more machine learning models which may be used to adapt control of powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, or another component. For example, data captured by one or more thermal sensors, first energy delivery device **16**, second energy delivery device **17**, MMD **19**, PFMS **18**, topology sensor **48**, mass source **44**, or other component may be used to train the machine learning device.

[0064] FIG. **2** is a conceptual and schematic diagram illustrating example additive manufacturing system **100**. Additive manufacturing system **100** of FIG. **2** may be an example of additive manufacturing system **10** of FIG. **1**. System **100** includes an example powder flow monitoring system **50** configured to monitor powder flow between a powder delivery device **52** and a build surface (not shown in FIG. **2**) during an additive manufacturing technique. Powder delivery device **52** may be an example of powder delivery device **14** of FIG. **1**, and PFMS may be an example of PFMS **18** of FIG. **1**.

[0065] Powder delivery device **52** includes a deposition head **54** that carries a plurality of powder nozzles **56**. Plurality of powder nozzles **56** output a powder stream **58** toward the build surface. As shown in FIG. **2**, the powder stream **58** may be focused at a focal plane, such that powder stream **58** is converging toward the focal plane and diverging away from the focal plane. As discussed above, deposition head **54** may further include a primary energy delivery device, which is not illustrated in FIG. **2** for improved clarity.

[0066] PFMS **18** includes a housing **60** (also referred to as an enclosure), which encloses an imaging device **62** and an illumination device **64**. In some examples, imaging device **62** may be a high-speed camera and illumination device **64** may be laser illuminator. Housing **60** is attached to an adjustable z-stage **66** by a bracket **68**.

[0067] Housing **60** may enclose imaging device **62** and illumination device **64** and help protect imaging device **62** and illumination device **64** from a surrounding environment. For instance, housing **60** may surround imaging device **62** and illumination device **64** and prevent any powder that reflects from the build surface toward PFMS **18** from impacting imaging device **62** or illumination device **64**.

[0068] Further, housing **60** may be configured to cool imaging device **62** and illumination device **64**. Imaging device **62** and illumination device **64** may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device **62** and illumination device **64** may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS **50** may include a cooling system **70** that removes heat from within housing **60** to cooling imaging device **62** and illumination device **64**. For instance, cooling system **70** may include cooling fluid circuit through which a cooling fluid flows, and housing **60** may include part of the cooling circuit. In some examples, housing **60** may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing **60** to cooling system **70** (e.g., a cooling fluid flowing through cooling system **70**).

[0069] As described above, PFMS **50** may be configured to measure powder flow of powder stream **58** (FIG. **2**) at one or more axial (or longitudinal) locations of powder stream **58** and determine one or more parameters associated with the powder flow. For instance, illumination device **64** may illuminate powder of powder stream **58** in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device **52** to the build surface. PFMS **50** may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device **52** and the build surface. Imaging device **62** may be configured to image at least some of the illuminated powder.

[0070] Included in the illustration of system **100** of FIG. **2** are secondary energy delivery device **74**, MMD **78**, and tertiary energy delivery device **76**. Secondary energy delivery device **74** may be a laser configured to deliver energy locally to a portion of the component. Tertiary energy delivery device **76**, in the illustrated example, is an induction heater configured to add thermal energy to the entire component. It is also considered that tertiary energy delivery **76** could be another type of heater, or even be a cooling device configured to remove thermal energy from the build surface of the component. In such examples, tertiary energy delivery device **76** may include a heat exchanger. System **100** includes MMD **78**. In the illustrated example of FIG. **2**, MMD **78** includes an ultrasound probe configured to direct acoustic energy **79** toward the additively manufactured component and receive acoustic energy reflected back from the component for detection and analysis. Each of secondary energy delivery device **74**, MMD **78**, and tertiary energy delivery device **76** may be communicatively coupled to and under control of a computing device (**12**, FIG. **1**).

[0071] FIG. **3** is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system. As shown in FIG. **3**, since powder is flowing in powder stream **58** at a relatively high velocity, imaging device **62** may not capture images of all the powder in powder stream **58**. The fraction of powder that imaging device **62** captures images of may be a function of average powder velocity at the image plane and a frame rate or capture speed of imaging device **62**. This is represented in FIG. **3** as “sampled” particles and “missed population” particles. The fraction of particles imaged by imaging device **62** may, in some examples, be less than about 50%, less than about 40%, less than about 30%, less than about 25%, less than about 20%, or less than about 15%.

[0072] PFMS **50** may include a computing device (e.g., computing device **12** of FIG. **1**) configured to analyze images captured by imaging device **62** to identify a number of particle detections in each captured image and, optionally, derive further parameters from the number of particle detections. As such, computing device **12** may receive image data representing an image captured by imaging device **62**. The image data may include representations of illuminated powder of powder stream **58**, as imaged by imaging device **62** (e.g., as captured in an image frame by imaging device **62**). Computing device **12** may generate a representation of powder stream based on the image data and output the representation of the powder stream for display at a display device.

[0073] For instance, computing device **12** may determine a powder mass flow represented by the image data. To do so, computing device **12** may identify a number of powder particles within each image frame. In some examples, computing device **12** additionally may identify a size and/or shape of each powder particle within each image frame. Computing device **12** may implement any suitable image analysis technique to identify powder particles, and, optionally, size and/or shape of powder particles.

[0074] Once computing device **12** has identified a number of powder particles within an image frame, computing device **12** may determine a mass flow based on the number of powder particles. For example, computing device **12** may determine the mass flow based on a calibration equation or calibration curve. FIG. **5** is an example calibration curve of particle detections versus mass flow. As shown in FIG. **4**, the relationship between particle detections may be substantially linear.

[0075] The relationship between particle detections and mass flow may be determined experimentally. For instance, the relationship between particle detections and mass flow may be determined for each powder type (e.g., composition, size distribution, or both), as each powder type may have a different relationship between particle detections and mass flow. The relationship may be determined experimentally by flowing a known mass of powder at a known rate, and imaging the powder. By doing this multiple times at multiple rates, the calibration curve may be generated. The curve, in the form of an equation, a look-up table, or the like, may be stored in computing device **12**, and computing device **12** may use the calibration curve to determine mass flow of a similar type of powder at a different flow rate based on particle detections.

[0076] In some examples, computing device **12** may receive image data representative of a sequence of images of illuminated powder in powder stream **58**. Each image may be associated with a time. As such, computing device **12** may select one or more images of the sequence of images and analyze the one or more images. For each selected image, computing device **12** may identify a number of particle detections and, optionally, determine a mass flow associated with powder stream **58** for each image frame.

[0077] As described above, system **10** may include both mass flow monitoring and heat flow monitoring. FIG. **1** best illustrates the mass flow monitoring aspects of system **10**. FIG. **5** is a conceptual block diagram illustrating further aspects of system **10**, best illustrating the heat flow monitoring aspects of the example system. System **10** includes an optical system **80** for observing thermal emissions around melt pool **32** and a melt pool monitor **15** including a thermal camera for monitoring a size and/or temperature of melt pool **32**. Identical reference numerals in FIGS. **1** and **5** refer to the same parts. Further, those common parts are the same or substantially identical, aside from any differences described herein.

[0078] As shown in FIG. **5**, primary energy delivery device **16** includes an optical system **80**. Although optical system **80** is shown and described only as associated with primary energy delivery device **16**, secondary energy delivery device **17** may include a corresponding optical system. During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Primary energy delivery device **16** may direct energy **34** at portion P**1** of first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0079] In some examples, temperature probe **21** may measure a temperature of build surface **28** in portion P**2** and output captured temperature information to computing device **12**. Although described and illustrated as a temperature probe herein, element **21** may be more generally referred to as thermal sensor **21**, and is not necessarily a temperature probe. For example, thermal sensor **21** may be configured to sense heat via an optical system, as described elsewhere herein. Accordingly, system **10** may include one or more thermal sensors (e.g., a single thermal camera of optical system **80**, or multiple thermal cameras, or multiple temperature probes, or combinations of thermal cameras and temperature probes, or the like). In any event, system **10** includes one or more thermal sensors configured to capture data indicative of a temperature of first portion P**1** of component **22** and second portion P**2** of component **22**.

[0080] In some examples, different portions of component **22** may be defined relative to melt pool **32**. In such examples, temperature probe **21** may be movable relative to build surface **28** to maintain a spatial relationship to melt pool **32**. In such examples, first portion P**1** may be defined as encompassing the portion of build surface **28** that defines melt pool **32** (and/or melt pool **32** plus the area within 1.5 radii of melt pool **32**, 2.0 radii of melt pool **32**, or 3.0 radii of melt pool **32**). In such examples. Second portion P**2** may be defined as encompassing an area with a radius equal to melt pool **32** but trailing melt pool **32** along build surface **28** relative to a toolpath by, for example, two, four, or six radii of melt pool **32**. Other portions (P**3**, P**4**, etc.) may be defined leading melt pool **32**, orthogonal to melt pool **32**, etc., and each portion may have a corresponding temperature probe **21**.

[0081] Alternatively, portions (P**1**, P**2**, etc.) may be defined relative to the expected final dimensions of component **22**. In such examples, probe **21** may be stationary or may be configured to vertically adjust as layers are added to measure a temperature of build surface **28** in same or similar location. Primary energy delivery device **16** may travel through portion P**2**, forming melt pool **32** as it travels along a toolpath. In such examples, computing device **12** may store instructions for energy **34** applied to each portion (P**1**, P**2**, etc.) to reduce or eliminate residual stresses. Computing device **12** may modify energy **34** delivered by primary energy delivery device **16**, secondary energy delivery device **17** or a tertiary energy delivery device (or cooling device) to conform to the stored instructions. In some examples, a plurality of portions of component **22** may be divided into a matrix with each portion of the plurality of portions corresponding to a cell in the matrix. In some examples, computing device **12** may similarly store instruction for each portion of the plurality of portions. In some examples, component **22** may be divided into a matrix comprising tens of individual portions, or hundreds of individual portions, or thousands of individual portions.

[0082] Computing device **12** may control secondary energy delivery device **17** to modify the temperature and cooling rate of portion P**2** of the build surface **28** by adding energy **34** to portion P**2** to ensure that component P**2** cools according to the desired thermal history profile, such that resulting component **22** has the desired thermal history relative to other portions of the component so that residual stresses are not created as the forces from differential thermal expansion and contraction across the different portions of the component. In some examples portions P**1** and P**2** may receive the same total amount of energy **34**, or may receive different amounts of energy **34** by primary energy source **16**, secondary energy source **17**, or both. Although portions PI and P**2** are described and illustrated as displaced from each other in the X-Y plane, it should also be considered that layer **24** and **26** may be considered different portions of component **22** in some cases. In such examples, portions P**1** and P**2** may be displaced from each other in the Z-direction. It should be noted that computing device **12** may control primary energy delivery device and secondary energy delivery device **17** independently based on data from topology sensor **48**, MP monitor **15**, optical system **80**, probe **21**, or combinations thereof. For example, computing device **12** may cause secondary energy delivery device **17** to deliver energy **34** to build surface **28** before, during, and/or subsequent to primary energy delivery device **16** delivering energy **34**.

[0083] In many cases, energy **34** output by primary energy delivery device **16** and/or secondary energy delivery device **17** is very high temperature and the intensity of its thermal emissions is significantly greater than the intensity of thermal emissions from melt pool **32**, and the surrounding areas of component **22**. Similarly, thermal emissions intensity at and near the center of melt pool **32** may be significantly greater than the intensity of thermal emissions near the edge of melt pool **32** and in areas surrounding melt pool **32**. Because of this, it may be difficult to accurately measure temperature and cooling rate of areas near the edge of melt pool **32** and in areas surrounding melt pool **32**. This results in difficulty predicting and controlling microstructure of the additively manufactured component **22**.

[0084] Optical system **80** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **80** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range.

[0085] The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device. In some examples, at least a portion of the optical train is coaxial with the axis at which energy delivery device **16** outputs energy, and the at least a portion of the optical train may be attached to or otherwise configured to move with the portion of energy delivery device **16** that directs or focuses energy **34** at or near the surface of component **22**. In this way, optical system **80** may move with energy delivery device **16** and track melt pool **32** as melt pool **32** moves across component **22**, without needing to correct for any offsets between energy delivery device **16** and optical system **80** and/or needing to correct for geometry of component **22**. In other examples, the optical train may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** may be configured to compensate for the offset and any affects this may have on the imaging, including shadowing, interference, geometry of component **22**, or the like.

[0086] Optical system **80** may include an occulting device. The occulting device is configured to reduce or block emissions (e.g., thermal emissions) that originate from the energy output by energy delivery device **16** and/or near a center of melt pool **32**, which otherwise obfuscate emissions from solidifying regions of material at or near the edge of melt pool **32** and outside of melt pool **32**. The occulting device may be a rigid occulting device or a dynamic occulting device. A rigid occulting device reduces or blocks emissions from a fixed region, e.g., from the energy **34** output by energy delivery device **16**. For instance, a rigid occulting device may include a device with fixed dimensions that is opaque to wavelengths of interest. As another example, a rigid occulting device may include an apodizing lens in which a center of the lens if substantially opaque to wavelengths of interest and opacity decreases as a function of radius.

[0087] A dynamic occulting device is configured to be controlled to occult different regions, e.g., different sizes and/or shapes. A dynamic occulting device may include a rigid occulting device that is mounted to a device that can translate the rigid occulting device along and/or perpendicularly to the optical axis. As another example, a dynamic occulting device may include an opaque and viscous liquid, such as mercury, contained between two substrates. The substrates are substantially transparent to the wavelength(s) of interest. One or both of the substrates may be movable relative to the other substrate to control the distance between the substrates. By reducing the distance between the substrates, the size of the occulting region may increase. By increasing the distance between the substrates, the size of the occulting region may decrease. As a third example, a dynamic occulting device may include a digital micromirror device. Computing device **12** may control the micromirrors of the digital micromirror device to direct emissions that originate from energy **34** output by energy delivery device **16** and/or near a center of the melt pool away from the imaging device. A digital micromirror device may enable control of both the size and shape of the region of emissions that are occulted.

[0088] FIG. **6** is a conceptual block diagram illustrating an example optical system **80** for observing thermal emissions at and/or around a melt pool **32** and or a focused spot on a build surface **28** where a melt pool is not formed (e.g., by second energy delivery device **17**, FIGS. **1** and **5**) formed during an additive manufacturing technique. Optical system **80** includes an optical train that includes first imaging optics **92**, occulting device **94**, second imaging optics **96**, and imaging device **98**. Imaging device **98** may be any suitable imaging device, including, for example, a visible light imaging device, an infrared imaging device, an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range, a two color pyrometry imaging device, or the like.

[0089] First and second imaging optics **92** and **96** may each include one or more optical devices used to direct light to imaging device **98**. For example, First and second imaging optics **92** and **96** may each include one or more refractive optical device (e.g., a lens), one or more reflective optical device (e.g., a mirror), one or more diffractive optical devices (e.g., a grating), one or more dichroic optical devices (e.g., a dichroic filter or mirror), or the like. Although two sets of imaging optics **92** and **98** are shown in FIG. **2**, in other examples, system **80** may include a single set of imaging optics or more than two sets of imaging optics.

[0090] Occulting device **94** is positioned within the optical train between first imaging optics **92** and second imaging optics **96**. In other example, occulting device **94** may be positioned between imaging device **98** and imaging optics **96** or after before imaging optics **92**. In some examples, occulting device **94** is positioned as the optical component nearest imaging device. This effectively results in removal of the portion of the image which occulting device **94** blocks. In other examples, occulting device **94** is positioned at another position within the optical train **80** where the image of component **22** resolves. Imaging optics **96** then may be configured to image occulting device **94** onto imaging device **98**.

[0091] As shown in FIG. **6**, in some examples, at least a portion of optical system **80** is coaxial with the axis at which energy delivery device **16** outputs energy **34** (i.e., a central longitudinal axis). For example, at least a portion of second imaging optics **92** (e.g., the portion at which thermal emissions **104** is incident upon second imaging optics **92**) may be coaxial with the axis at which energy delivery device **16** outputs energy **34**. This may reduce image manipulation that otherwise may be applied to the resulting image to correct for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like. In other examples, optical system **80** (e.g., the portion at which thermal emissions **104** are incident upon second imaging optics **92**) may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** (FIG. **1**) or another computing device may manipulate the resulting image to compensate for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like.

[0092] FIG. **6** also illustrates energy delivery device **16** outputting energy **34**, which is incident upon component **22** and results in formation of melt pool **32**. Surrounding melt pool is a cooling zone **102**, in which temperature gradients from the temperature of melt pool **32** to ambient temperature are present. As shown in FIG. **6**, melt pool **32** and cooling zone **102** emit thermal emissions **104** (e.g., thermal radiation), which travel through optical system **80** to imaging device **98**, which images the thermal emissions **104**. Occulting device **94** occults (e.g., reduces the intensity of or substantially eliminates) thermal emissions **104** from a selected region, e.g., a region corresponding to energy **34** and at least a portion of melt pool **32**. This may allow imaging device **98** to more effectively image relatively lower intensity thermal emissions from at or near the edge of melt pool **32** and within cooling zone **102**. This may enable more accurate measurement of temperatures within the cooling zone **102**, and heat flow within cooling zone **102**.

[0093] Returning to FIG. **5**, system **10** also includes melt pool monitor ("MP monitor") **15**. Melt pool monitor **15** may include a sensor for monitoring a characteristic of melt pool **32**. The monitored characteristic may be indicative of a temperature of melt pool **32**. For example, the sensor may include an imaging system, such as a visual or thermal camera, e.g., camera to visible light or infrared (IR) radiation. A visible light camera may monitor the geometry of the melt pool, e.g., a width, diameter, shape, or the like. A thermal (or IR) camera may be used to detect the size, temperature, or both of the melt pool. In some examples, a thermal camera may be used to detect the temperature of the melt pool at multiple positions within the melt pool, such as a leading edge, a center, and a trailing edge of the melt pool. In some examples, the imaging system may include a relatively high-speed camera capable of capturing image data at a rate of tens or hundreds of frames per second or more, which may facilitate real-time detection of the characteristic of the melt pool. In some examples, MP monitor **15** may capture data at a sequence of particular points in time including a first point in time, a second point in time, etc.

[0094] FIG. **7** is a process flow diagram illustrating an additive manufacturing monitor and control technique. The technique of FIG. **7** may be implemented by system **10** of FIGS. **1** and **5** and will be described with concurrent reference to FIGS. **1** and **5**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. **7** may be performed by other systems. For example, system **100** of FIG. **2** may perform the described technique.

[0095] One or more computing devices **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. **7**). For instance, one or more computing devices **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0096] One or more computing devices **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. Data received from powder flow mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Data from topology sensor **48** indicates powder mass captured by melt pool **32** and added to component **22**.

[0097] One or more computing devices **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**.

[0098] Further, one or more computing devices **12** may determine an overall mass flux using the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. **7**).

[0099] Similarly, one or more computing devices **12** may be configured to control first energy delivery device **16** and second energy delivery device **17** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. **7**). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16** and/or second energy delivery device **17**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**). In some examples, computing device **12** may be configured to operate one of the energy delivery devices (e.g., first energy delivery device **16**) at a constant heat input and to control second energy delivery device **17** to modify (e.g., selectively tailor) residual stress of a first portion P**1** and/or a second portion P**2**.

[0100] One or more computing devices **12** may be configured to receive data captured by one or more thermal sensors, such as optical system **80** and/or melt pool monitor **82**. One or more computing devices may determine a cooling rate and associated heat from using data from optical system **80** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor **15**, probe **21**, or both. One or more computing devices **12** may be configured to capture data from melt pool monitor **15**, probe **21** throughout a period of time (e.g., at a first point in time, a second point in time, a third point in time, etc.). One or more computing devices **12** may be configured to determine a residual stress of portions P**1**, P**2** of component **22** based on the captured data, and thus residual stresses throughout the entirety of component **22**. One or more computing devices **12** then may use the determined residual stress as an input to the control algorithm used to control first energy delivery device **16**, second energy delivery device **17**, or both. (see left of FIG. **7**).

[0101] In some examples, optical system **80** and/or probe **21** may be configured to sense, capture, and output to one or more computing device **12** data representative of a temperature of portions P**1**, P**2** of component **22**. One or more computing devices **12** may be configured to determine residual stresses of portions P**1**, P**2** based on the captured data. One or more computing devices **12** may then use the determined residual stresses to predict final dimensions of component **22**. In some examples, the predicted final dimensions may be based on the build strategy to build as-deposited component **22** and on the expected deformation of component **22** due to the determined residual stress.

[0102] In some examples, the predicted final dimensions of component **22** may be used as an input to a control algorithm of computing device **12** used to control first energy delivery device **16**, second energy delivery device **17**, or both. In some examples, for example if the predicted final dimensions of component **12** meet a threshold level of matching planned (e.g., technically specified) component dimensions, one or more computing devices **12** may be configured to continue the build based on the validated predicted final component dimensions. In some examples, if one or more computing devices **12** determines that the predicted final component dimensions do not meet a threshold level of matching planned component dimensions, one or more computing devices may be configured to cause system **10** to stop before building a flawed component **22**, or may be configured to output to a display a warning or alarm indicating improper residual stresses contained in component **22**. In some examples, one or more computing devices **12** also may use the deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the residual stresses, as the added powder mass and quench effects associated with the captured powder affect the cooling rate of component **22**.

[0103] FIGS. **8-10** are conceptual and schematic diagram illustrating an additively-manufactured component. Similar reference numerals in FIGS. **8-10** indicate similar elements System **10** of FIGS. **1** and **5** or system **100** of FIG. **2** may be used to additively manufacture any of the illustrated components.

[0104] FIG. **8** illustrates component **222** in an initial as-deposited state. Component **222** includes a layer-by-layer built component body **225**, and thus includes layers **224**A, **224**B, and **224**C. In the illustrated example, the planned final dimensions of component **222** include overall length L.sub.0 and overall height H.sub.0. Although illustrated as a simple block shape having a length and height in the illustrated example, it is considered that component **222** may define a complex shape in other examples, and thus have complex dimensions mapped in a 3D space model.

[0105] The initial build strategy, that is, the settings of the additive manufacturing system of component **222** during manufacturing (e.g., system **10**), may not include accounting for residual stresses within component **222** that may cause deformation such that the as-deposited dimensions fail to match the planned final dimensions. As such, in this example, layers **224**A, **224**B, and **224**C are deposited at heights H.sub.1, H.sub.2, and H.sub.3 uniformly to achieve overall height H.sub.0, which is the planned overall height of component **222**. Component **222** defines portions P**1**, P**2**, and P**3**. Portions P**1**, P**2**, and P**3** define lengths L.sub.1, L.sub.2, and L.sub.3 respectively to define overall length L.sub.0. Portions P**1** and P**3** are defined as near edges of component **222**, only one edge of which is labeled as edge **229** in FIG. **8**. Portion P**2** is defined in a central portion of component **222** away from edges of component **222**.

[0106] FIG. **9** is a conceptual and schematic diagram illustrating component **222** in a final state. In some examples, the final state may be achieved after component **222** is removed from a mechanically supporting stage and ready for operational use. As component **222** ages and cools from the as-deposited state of FIG. **8** to the final state of FIG. **9**, thermal energy **231** (illustrated by block arrows) may diffuses from component **222** into the surrounding environment. Portions P**1** and P**3** may, due to the close proximity and increased surface area with the environment, cool more quickly than central portion P**2**, causing deformation of component **222**, illustrated by the bowing in FIG. **9**. Thus, FIGS. **8** and **9** illustrate one potential problem with residual stresses in component **222**. Although the residual stresses causing deformation are described as stemming from forces of differential thermal contraction of portions P**1**, P**2**, and P**3** relative to neighboring portions, other causes of residual stress may be identified by computing device **12**.

[0107] FIG. **10** is a conceptual and schematic diagram illustrating component **322** in an initial as-deposited state, accounting for predicted residual stresses in the component. In the illustrated example, the planned final component dimensions are those in FIG. **8**. The as-deposited component of FIG. **10** includes layers **324**A, **324**B, and **324**C, and defines portions P**1**, P**2**, and P**3**. With concurrent reference to FIGS. **1** and **8-10**, computing device **12** of system **10** may modify the build strategy of component **322** after determining thermal stresses in component **322** and predicting that the as-deposited dimensions of FIG. **8** will result in the final dimensions of FIG. **9** after exertion of the thermal stresses. Responsive to predicting final component dimensions that do not meet a threshold for matching planned component dimensions, computing device **12** may modify the build strategy to counteract the thermal stresses and adjust the predicted final dimensions to better match the planned final dimensions. In the illustrated example of FIG. **10**, computing device **12** modifies settings of a mass flux module, although the build strategy may, in other examples, include modification of a heat flux module, alone or in combination with changes to mass flux.

[0108] For example, computing device **12** may cause powder delivery device **14** to modify mass flow of powder in powder stream **30** to apply more mass of powder to portions PI and P**3**, which are adjacent to the edges of component **329**, to counteract and/or correct for the predicted effects of residual stresses based on data captured by one or more thermal sensors. Although illustrated in FIG. **10** as modifying powder delivery device **14** to adjust the mass flow rate of powder in powder stream **30** for ease of illustration, in some examples, modification of energy **34** delivered by first energy delivery device **16** or second energy delivery device **17** to create component **322** with final actual dimensions that match the planned dimensions.

[0109] FIG. **11** is a flow diagram illustrating an example technique for managing residual stresses in an additively-manufactured component according to the present disclosure. The illustrated technique may be performed by system **10** of FIGS. **1** and **5**, and will be described as performed by system **10**, but other systems may be used to perform the described techniques, such as system **100** of FIG. **2**. Furthermore, system **10** and system **100** may be used to perform other techniques.

[0110] Computing device **12** may receive data indicative of the temperature of first portion P**1** of component from one or more thermal sensors **21** of system **10** captured at a first point in time and at a second point in time (**402**). In some examples, system **10** may include first energy delivery device **16** configured to deliver energy **34** to build surface **28** of component **22** to form melt pool **32** in build surface **2**. In some examples, system **10** may include second energy delivery device **17** configured to deliver energy to build surface **28** of the component **22**. System **10** also includes powder delivery device **14** configured to direct powder stream **30** toward melt pool **32**. Computing device **12** may receive data indicative of a temperature of second portion P**2** from the one or more thermal sensors **21** at a second point in time (**404**).

[0111] Computing device **12** may determin a residual stress of component **22** based at least partially on the received thermal sensor data from first portion P**1** of component **22** and received data from second portion P**2** of component **22** (**406**). In some examples, determining the residual stress may include comprises calculating, by computing device **12**, a thermal expansion or contraction of the component **22** based at least partially on the data captured by one or more thermal sensors **21**. In some examples, calculating a thermal expansion or contraction may include calculating a cooling rate of first portion P**1** or second portion P**2** of component **22** by comparing the data captured at the first point in time to the data captured at the second point in time.

[0112] Computing device **12** may predict final dimensions of component **22** based at least partially on the determined residual stress of component **22** (**408**). In some examples, the technique of FIG. **11** may further include controlling, by computing device **12**, at least one of the powder delivery device **14**, first energy delivery device **16**, or second energy delivery device **17** based at least partially on predicted final dimensions of component **22** to adjust a build strategy of the component based on the determined residual stress. In some examples, the build strategy may include all settings of system **10** that result in the final dimensions of component **22**. As one example, the build strategy may be adjusted by causing secondary energy device **17** to apply energy **34** to portions of component **22** located at or near an edge of component **22** to equalize the cooling rate of one or more edge portions with the cooling rate of one or more central portions to reduce stresses induced by different rates of thermal contraction of the edge portion or portions with the central portion or portions.

[0113] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0114] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0115] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory

(EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0116] In some examples, a computer-readable storage medium may include a non-transitory medium. The term "non-transitory" may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0117] Various examples have been described. These and other examples are within the scope of the following examples and claims.

[0118] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; one or more thermal sensors configured to measure a temperature of a first portion of the additively-manufactured component and a second portion of the additively manufactured component; and a computing device configured to: receive data indicative of the temperature of the first portion of the additively-manufactured component from the one or more thermal sensors captured at a first point in time and data indicative of the temperature of the first portion captured at a second point in time; receive data indicative of a temperature of the second portion of the additively-manufactured component from the one or more thermal sensors captured at the first point in time and data indicative of a temperature of the second portion captured at the second point in time; determine a residual stress of the additively-manufactured component based at least partially on the received data indicative of the temperature of the first portion of the additively-manufactured component and the received data indicative of the temperature of the second portion of the additively-manufactured component; and predict final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.

[0119] Example 2: The additive manufacturing system of example 1, further comprising a second energy delivery energy delivery device configured to deliver energy to the build surface of the additively-manufactured component.

[0120] Example 3: The additive manufacturing system of example 2, wherein the computing device is configured to mitigate deformation of the additively-manufactured component, where deformation is defined as the difference in one or more dimensions of the component between an as-deposited state and a final state, by controlling at least one of the first energy delivery device, the second energy delivery device, or the powder delivery device to reduce or eliminate residual stress in the additively-manufactured component.

[0121] Example 4: The additive manufacturing system of any of examples 2 and 3, wherein the computing device is configured to control at least one of the first energy delivery device, the second energy delivery device, or the powder delivery device to impart residual stress in the component to leverage distortion to achieve a final component dimension.

[0122] Example 5: The additive manufacturing system of any of examples 2 through 4,wherein: the computing device stores instructions and is configured to execute an initial build strategy, the computing device is further configured to modify the initial build strategy to form a modified build strategy based on the determined residual stress, and the computing device is further configured to control at least one of the powder delivery device, the first energy delivery device, or the second energy delivery device based at least a build strategy, and wherein the computing device is configured to adjust the build strategy of the component based on the determined residual stress.

[0123] Example 6: The additive manufacturing system of any of examples 1 through 5, wherein, to determine the residual stress, the computing device is configured to calculate thermal expansion or contraction of the additively manufactured component based at least partially on the data captured by the one or more thermal sensors.

[0124] Example 7: The additive manufacturing system of example 6, wherein the computing device is configured to calculate a cooling rate of the first portion or the second portion of the additively-manufactured component by comparing the data captured at the first point in time to the data captured at the second point in time.

[0125] Example 8: The additive manufacturing system of example 7, wherein the computing device is configured to compare the cooling rate of the first portion to the cooling rate of the second portion to determine the residual stress of the additively-manufactured component.

[0126] Example 9: The additive manufacturing system of any of examples 7 and 8, wherein, to determine the residual stress, the computing device is configured to receive data indicative of a temperature of a plurality of portions of the additively-manufactured component in addition to the first portion and the second portion.

[0127] Example 10: The additive manufacturing system of any of examples 1 through 9, further comprising a stage configured to mechanically support the additively-manufactured component, and wherein the computing device is configured to determine the residual stress of the additively-manufactured component without removing the additively-manufactured component from the stage.

[0128] Example 11: The additive manufacturing system of any of examples 1 through 10, further comprising a plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system.

[0129] Example 12: The additive manufacturing system of any of examples 2 through 11, wherein: the first energy delivery device is coincident with a central longitudinal axis of a deposition head, and the second energy delivery device is not coincident with the central longitudinal axis of the deposition head.

[0130] Example 13: The additive manufacturing system of any of examples 1 through 12, wherein the computing device is configured to control the energy delivery device based on the determined residual stress by: modifying at least one of a power, a travel speed, a spot size, or a power density of the first energy delivery device or the second energy delivery device, or modifying a mass flow rate of the powder to adjust the as-deposited thickness of a layer being added to the additively-manufactured component.

[0131] Example 14: The additive manufacturing system of any of examples 1 through 13, wherein the computing device stores and is configured to execute a machine learning algorithm, and wherein the machine learning algorithm is trained on data generated by the one or more thermal sensors.

[0132] Example 15: The additive manufacturing system of any of examples 1 through 14, further comprising a topology sensor configured to capture data indicative of a topology of the build surface, and wherein the computing device is configured to receive data generated by the topology sensor and determine the residual stress based at least partially on the data generated by the topology sensor.

[0133] Example 16: A method includes receiving, by a computing device, data indicative of a temperature of a first portion of an additively-manufactured component from one or more thermal sensors of an additive manufacturing system captured at a first point in time and at a second point in time; receiving, by the computing device, data indicative of a temperature of a second portion of the additively-manufactured component from the one or more thermal sensors at a second point in time; determining, by the computing device, a residual stress of the additively-manufactured component based at least partially on the received data indicative of the temperature of the first portion of the additively-manufactured component and the received data indicative of the temperature of the second portion of the additively-manufactured component; and predicting, by the computing device, final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.

[0134] Example 17: The method of example 16, further comprising delivering energy to a build surface of the additively manufactured component via a second energy delivery energy delivery device.
[0135] Example 18: The method of example 17, further includes delivering, by a powder deliver device, a stream of powder to a melt pool in a build surface of the additively-manufactured component, and controlling, by the computing device, at least one of the powder delivery device, the first energy delivery device, or the second energy delivery device based at least partially on the predicted final dimensions of the additively-manufactured component to adjust a build strategy of the component based on the determined residual stress, wherein the build strategy results in the final dimensions of the additively-manufactured component.
[0136] Example 19: The method of any of examples 16 through 18, wherein determining the residual stress comprises calculating, by the computing device, a thermal expansion or contraction of the additively manufactured component based at least partially on the data captured by the one or more thermal sensors.
[0137] Example 20: The method of example 19, wherein calculating a thermal expansion or contraction comprises calculating a cooling rate of the first portion or the second portion of the additively-manufactured component by comparing the data captured at the first point in time to the data captured at the second point in time.

## Claims

**1**. An additive manufacturing system comprising: an energy delivery device configured to deliver energy to a build surface of an additively-manufactured component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; one or more thermal sensors configured to measure a temperature of a first portion of the additively-manufactured component and a second portion of the additively manufactured component; and a computing device configured to: receive data indicative of the temperature of the first portion of the additively-manufactured component from the one or more thermal sensors captured at a first point in time and data indicative of the temperature of the first portion captured at a second point in time; receive data indicative of a temperature of the second portion of the additively-manufactured component from the one or more thermal sensors captured at the first point in time and data indicative of a temperature of the second portion captured at the second point in time; determine a residual stress of the additively-manufactured component based at least partially on the received data indicative of the temperature of the first portion of the additively-manufactured component and the received data indicative of the temperature of the second portion of the additively-manufactured component; and predict final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.
**2**. The additive manufacturing system of claim 1, further comprising a second energy delivery energy delivery device configured to deliver energy to the build surface of the additively-manufactured component.
**3**. The additive manufacturing system of claim 2, wherein the computing device is configured to mitigate deformation of the additively-manufactured component, where deformation is defined as the difference in one or more dimensions of the component between an as-deposited state and a final state, by controlling at least one of the first energy delivery device, the second energy delivery device, or the powder delivery device to reduce or eliminate residual stress in the additively-manufactured component.
**4**. The additive manufacturing system of claim 2, wherein the computing device is configured to control at least one of the first energy delivery device, the second energy delivery device, or the powder delivery device to impart residual stress in the component to leverage distortion to achieve

a final component dimension.

**5**. The additive manufacturing system of claim 2, wherein: the computing device stores instructions and is configured to execute an initial build strategy, the computing device is further configured to modify the initial build strategy to form a modified build strategy based on the determined residual stress, and the computing device is further configured to control at least one of the powder delivery device, the first energy delivery device, or the second energy delivery device based at least a build strategy, and wherein the computing device is configured to adjust the build strategy of the component based on the determined residual stress.

**6**. The additive manufacturing system of claim 1, wherein, to determine the residual stress, the computing device is configured to calculate thermal expansion or contraction of the additively manufactured component based at least partially on the data captured by the one or more thermal sensors.

**7**. The additive manufacturing system of claim 6, wherein the computing device is configured to calculate a cooling rate of the first portion or the second portion of the additively-manufactured component by comparing the data captured at the first point in time to the data captured at the second point in time.

**8**. The additive manufacturing system of claim 7, wherein the computing device is configured to compare the cooling rate of the first portion to the cooling rate of the second portion to determine the residual stress of the additively-manufactured component.

**9**. The additive manufacturing system of claim 7, wherein, to determine the residual stress, the computing device is configured to receive data indicative of a temperature of a plurality of portions of the additively-manufactured component in addition to the first portion and the second portion.

**10**. The additive manufacturing system of claim 1, further comprising a stage configured to mechanically support the additively-manufactured component, and wherein the computing device is configured to determine the residual stress of the additively-manufactured component without removing the additively-manufactured component from the stage.

**11**. The additive manufacturing system of claim 1, further comprising a plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system.

**12**. The additive manufacturing system of claim 2, wherein: the first energy delivery device is coincident with a central longitudinal axis of a deposition head, and the second energy delivery device is not coincident with the central longitudinal axis of the deposition head.

**13**. The additive manufacturing system of claim 1, wherein the computing device is configured to control the energy delivery device based on the determined residual stress by: modifying at least one of a power, a travel speed, a spot size, or a power density of the first energy delivery device or the second energy delivery device, or modifying a mass flow rate of the powder to adjust the as-deposited thickness of a layer being added to the additively-manufactured component.

**14**. The additive manufacturing system of claim 1, wherein the computing device stores and is configured to execute a machine learning algorithm, and wherein the machine learning algorithm is trained on data generated by the one or more thermal sensors.

**15**. The additive manufacturing system of claim 1, further comprising a topology sensor configured to capture data indicative of a topology of the build surface, and wherein the computing device is configured to receive data generated by the topology sensor and determine the residual stress based at least partially on the data generated by the topology sensor.

**16**. A method comprising: receiving, by a computing device, data indicative of a temperature of a first portion of an additively-manufactured component from one or more thermal sensors of an additive manufacturing system captured at a first point in time and at a second point in time; receiving, by the computing device, data indicative of a temperature of a second portion of the additively-manufactured component from the one or more thermal sensors at a second point in time; determining, by the computing device, a residual stress of the additively-manufactured component based at least partially on the received data indicative of the temperature of the first portion of the additively-manufactured component and the received data indicative of the temperature of the second portion of the additively-manufactured component; and predicting, by the computing device, final dimensions of the additively-manufactured component based at least partially on the determined residual stress of the additively-manufactured component.

**17**. The method of claim 16, further comprising delivering energy to a build surface of the additively manufactured component via a second energy delivery energy delivery device.

**18**. The method of claim 17, further comprising: delivering, by a powder deliver device, a stream of powder to a melt pool in a build surface of the additively-manufactured component, and controlling, by the computing device, at least one of the powder delivery device, the first energy delivery device, or the second energy delivery device based at least partially on the predicted final dimensions of the additively-manufactured component to adjust a build strategy of the component based on the determined residual stress, wherein the build strategy results in the final dimensions of the additively-manufactured component.

**19**. The method of claim 16, wherein determining the residual stress comprises calculating, by the computing device, a thermal expansion or contraction of the additively manufactured component based at least partially on the data captured by the one or more thermal sensors.

**20**. The method of claim 19, wherein calculating a thermal expansion or contraction comprises calculating a cooling rate of the first portion or the second portion of the additively-manufactured component by comparing the data captured at the first point in time to the data captured at the second point in time.